Solution

The length a of the semi major axis of the elliptical ceiling is $17m$. The height b of the semi minor axis is $8m$. Thus

$$c^2 = a^2 - b^2 = 17^2 - 8^2$$

$$\text{then} \quad c = \sqrt{289-64} = \sqrt{225}$$

$$\simeq 15$$

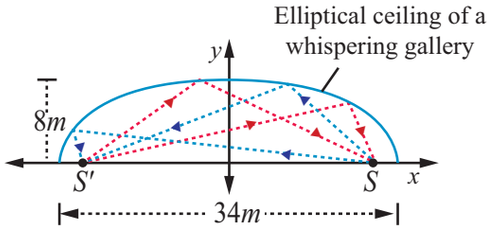


Fig. 5.64

For the elliptical ceiling the foci are located on either side about $15m$ from the centre, along its major axis.

A non-invasive medical miracle

In a lithotripter, a high-frequency sound wave is emitted from a source that is located at one of the foci of the ellipse. The patient is placed so that the kidney stone is located at the other focus of the ellipse.

Example 5.39

If the equation of the ellipse is $\frac{(x-11)^2}{484} + \frac{y^2}{64} = 1$ (x and y are measured in centimeters) where to the nearest centimeter, should the patient's kidney stone be placed so that the reflected sound hits the kidney stone?

Solution

The equation of the ellipse is $\frac{(x-11)^2}{484} + \frac{y^2}{64} = 1$. The origin of the sound wave and the kidney stone of patient should be at the foci in order to crush the stones.

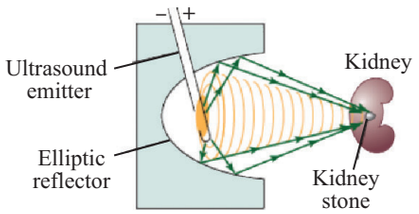


Fig. 5.65

$$a^2 = 484 \text{ and } b^2 = 64$$

$$c^2 = a^2 - b^2$$

$$= 484 - 64$$

$$= 420$$

$$c \simeq 20.5$$

Therefore the patient's kidney stone should be placed 20.5cm from the centre of the ellipse.

5.7.6 Reflective Property of a Hyperbola

The lines from the foci to a point on a hyperbola make equal angles with the tangent line at that point (Fig. 5.66).

The light or sound or radio waves directed from one focus is received at the other focus as in the case ellipse (Fig. 5.54) used in spotting location of ships sailing in deep sea.

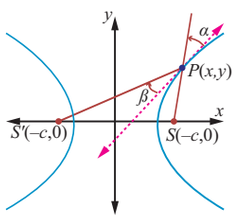


Fig. 5.66

Example 5.40

Two coast guard stations are located 600 km apart at points $A(0,0)$ and $B(0,600)$. A distress signal from a ship at P is received at slightly different times by two stations. It is determined that the ship is 200 km farther from station A than it is from station B. Determine the equation of hyperbola that passes through the location of the ship.

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